

ST MARY'S, AK, USA

WMO: 702005

Lat: 62.062N

Lon: 163.300W

Elev: 312

StdP: 14.53

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26647

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-17.6	-28.6	1.1	-16.4	-26.5	1.3	-14.0	35.7	17.8	33.7	16.8	6.0	40	0.855

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	71.7	58.0	66.4	56.2	63.6	54.4	59.8	67.7	57.7	64.5	55.9	62.0	9.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.3	68.2	61.6	54.5	63.9	59.7	53.0	60.3	58.5	26.2	68.2	24.9	64.5	23.8	61.9	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.9	26.6	23.9	DB	-25.4	75.2	6.5	4.5	-30.1	78.4	-33.9	81.0	-37.6	83.5	-42.3	86.7	(4)
(5)			WB	-23.4	62.3	5.2	3.5	-27.1	64.8	-30.1	66.9	-33.0	68.8	-36.8	71.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	7.6	12.1	11.8	26.0	40.4	50.8	53.7	51.6	44.3	30.4	18.2	9.7	
	DBStd	20.11	15.55	14.68	13.37	11.50	8.28	5.60	5.01	5.09	5.88	8.49	11.68	13.92	
	HDD50	7694	1313	1061	1186	719	312	54	18	38	181	608	955	1249	
	HDD65	12849	1778	1481	1651	1169	764	426	352	415	621	1073	1405	1714	
	CDD50	322	0	0	0	0	13	78	133	88	10	0	0	0	
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0	
	CDH74	36	0	0	0	0	1	8	19	8	0	0	0	0	
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0	
Wind	WSAvg	10.9	13.1	14.0	12.1	11.5	10.0	8.9	8.8	9.3	9.4	9.9	11.6	12.4	
Precipitation	PrecAvg	24.9	1.7	1.8	1.5	1.3	1.3	1.7	2.4	3.2	3.1	2.6	2.5	2.0	
	PrecMax	33.9	3.6	4.9	4.8	2.6	2.8	2.9	4.4	6.2	6.1	4.3	5.7	4.2	
	PrecMin	19.9	0.4	0.5	0.3	0.3	0.4	0.6	0.9	1.2	0.2	0.5	0.3	0.7	
	PrecStd	3.5	0.9	1.1	1.0	0.6	0.6	0.4	0.8	1.0	1.1	0.9	1.4	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	36.8	38.3	53.8	71.7	74.7	76.6	72.9	62.8	50.3	51.9	38.3	
		MCWB	33.8	33.9	34.5	43.0	52.8	58.1	62.9	61.6	52.5	45.4	42.3	33.2	
	2%	DB	35.6	35.0	36.0	47.7	63.2	70.3	71.6	66.3	57.4	46.6	42.5	35.5	
		MCWB	33.7	33.0	33.2	39.9	49.7	56.3	58.8	58.4	50.4	43.9	38.6	33.9	
	5%	DB	32.4	33.6	33.7	43.4	58.3	65.8	66.4	63.3	54.7	44.7	36.3	32.4	
		MCWB	31.4	32.0	31.8	37.2	47.2	54.3	57.3	55.7	49.8	42.1	34.5	31.5	
	10%	DB	28.4	31.1	30.2	39.3	54.3	63.0	63.5	60.5	52.4	41.7	34.0	29.0	
		MCWB	27.3	30.3	29.0	34.9	45.4	52.5	55.6	54.7	48.8	39.3	32.5	28.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	34.7	35.3	43.2	53.4	60.3	63.9	63.2	54.7	47.0	42.6	35.0
			MCDB	36.9	35.9	37.3	51.1	67.4	72.0	74.9	70.2	59.4	48.6	49.1	36.3
2%		WB	33.2	33.0	33.2	40.3	50.3	57.3	60.0	59.6	52.2	44.8	37.2	33.1	
		MCDB	35.0	34.3	34.9	46.8	61.3	66.3	67.8	66.1	56.0	46.4	39.8	34.8	
5%		WB	31.4	31.9	31.6	37.7	48.0	55.2	58.2	57.2	50.6	42.3	34.3	31.5	
		MCDB	32.5	33.0	33.3	43.2	57.3	64.1	65.2	61.9	54.0	44.4	36.4	32.6	
10%		WB	27.8	29.8	29.0	35.4	45.9	53.2	56.5	55.0	49.1	39.7	32.1	28.2	
		MCDB	28.5	30.8	30.1	39.3	53.4	61.4	62.7	58.8	52.3	42.1	34.0	28.8	
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.0	11.9	12.6	15.1	16.2	12.7	11.3	11.0	9.3	8.4	9.6	
		MCDBR	9.3	9.6	10.6	15.0	23.5	23.6	20.8	16.9	14.7	9.7	7.8	10.9	
		MCWBR	8.6	8.8	8.6	9.9	13.3	12.6	11.1	9.9	9.2	7.6	7.2	10.1	
	5% WB	MCDBR	9.4	9.6	9.7	14.5	22.1	20.8	18.3	14.7	12.2	8.8	7.0	10.7	
		MCWBR	8.8	9.0	8.3	9.8	12.7	11.5	10.4	9.1	8.3	7.4	7.0	10.2	
Clear-Sky Solar Irradiance	taub	0.210	0.270	0.276	0.345	0.371	0.365	0.365	0.382	0.324	0.287	0.230	0.197		
	taud	2.036	2.166	2.234	2.094	2.261	2.379	2.417	2.345	2.503	2.402	2.018	1.882		
	Ebn at Noon	154	219	267	264	268	272	269	252	251	217	135	86		
	Edh at Noon	11	21	28	40	37	34	32	32	23	17	11	7		
All-Sky Solar Radiation	RadAvg	97	319	897	1415	1613	1669	1352	1075	711	347	113	44		
	RadStd	12	38	79	108	152	171	109	135	63	29	18	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	+3.01	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page